Verilog code for D Flip Flop

D Flip-Flop is a fundamental component in digital logic circuits. Verilog code for D Flip Flop is presented in this project. There are two types of D Flip-Flops being implemented which are Rising-Edge D Flip Flop and Falling-Edge D Flip Flop.

Verilog code for Rising Edge D Flip Flop:

```
// FPGA projects using Verilog/ VHDL
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// Verilog code for D Flip FLop
// Verilog code for rising edge D flip flop
module RisingEdge_DFlipFlop(D,clk,Q);
input D; // Data input
input clk; // clock input
output Q; // output Q
always @(posedge clk)
begin
 Q <= D;
end
endmodule
```

Verilog code for Rising Edge D Flip-Flop with Synchronous Reset:

```
// FPGA projects using Verilog/ VHDL
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// Verilog code for D Flip FLop
// Verilog code for Rising edge D flip flop with Synchronous Reset input
module RisingEdge_DFlipFlop_SyncReset(D,clk,sync_reset,Q);
input D; // Data input
input clk; // clock input
input sync reset; // synchronous reset
output reg Q; // output Q
always @(posedge clk)
begin
 if(sync reset==1'b1)
  Q <= 1'b0;
 else
  Q <= D;
end
endmodule
```

Verilog code for Rising Edge D Flip-Flop with Asynchronous Reset High Level:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com
// Verilog code for D Flip FLop
// Verilog code for Rising edge D flip flop with Asynchronous Reset high
module RisingEdge_DFlipFlop_AsyncResetHigh(D,clk,async_reset,Q);
input D; // Data input
input clk; // clock input
input async_reset; // asynchronous reset high level
output reg Q; // output Q
always @(posedge clk or posedge async_reset)
begin
 if(async_reset==1'b1)
  Q <= 1'b0;
```

```
 else
  Q <= D;
end
endmodule
```

Verilog code for Rising Edge D Flip-Flop with Asynchronous Reset Low Level:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com
// Verilog code for D Flip FLop
// Verilog code for Rising edge D flip flop with Asynchronous Reset Low
module RisingEdge_DFlipFlop_AsyncResetLow(D,clk,async_reset,Q);
input D; // Data input
input clk; // clock input
input async_reset; // asynchronous reset low level
output reg Q; // output Q
always @(posedge clk or negedge async_reset)
begin
 if(async_reset==1'b0)
  Q <= 1'b0;
 else
  Q <= D;
end
endmodule
```

Verilog code for Falling Edge D Flip Flop:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com
// Verilog code for D Flip FLop
// Verilog code for falling edge D flip flop
module FallingEdge_DFlipFlop(D,clk,Q);
input D; // Data input
input clk; // clock input
output reg Q; // output Q
always @(negedge clk)
begin
 Q <= D;
end
endmodule
```

Verilog code for Falling Edge D Flip-Flop with Synchronous Reset:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com
// Verilog code for D Flip FLop
// Verilog code for Falling edge D flip flop with Synchronous Reset input
module FallingEdge_DFlipFlop_SyncReset(D,clk,sync_reset,Q);
input D; // Data input
input clk; // clock input
input sync_reset; // synchronous reset
output reg Q; // output Q
always @(negedge clk)
begin
 if(sync_reset==1'b1)
  Q <= 1'b0;
 else
  Q <= D;
end
endmodule
```

Verilog code for Falling Edge D Flip-Flop with Asynchronous Reset High Level:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com
// Verilog code for D Flip FLop
// Verilog code for Falling edge D flip flop with Asynchronous Reset high
module FallingEdge_DFlipFlop_AsyncResetHigh(D,clk,async_reset,Q);
input D; // Data input
input clk; // clock input
```

```
input async_reset; // asynchronous reset high level
output reg Q; // output Q
always @(negedge clk or posedge async_reset)
begin
 if(async_reset==1'b1)
  Q <= 1'b0;
 else
  Q <= D;
end
endmodule
```

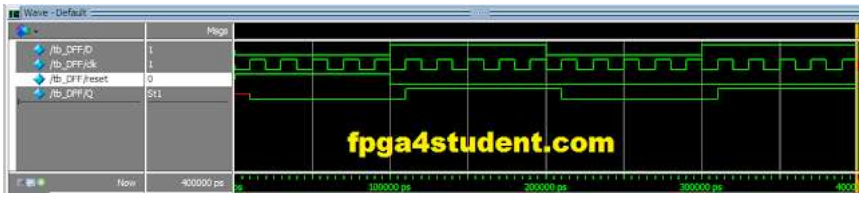
Verilog code for Falling Edge D Flip-Flop with Asynchronous Reset Low Level:

```
// FPGA projects using Verilog/ VHDL
// fpga4student.com
// Verilog code for D Flip FLop
// Verilog code for Falling edge D flip flop with Asynchronous Reset low
module FallingEdge_DFlipFlop_AsyncResetLow(D,clk,async_reset,Q);
input D; // Data input
input clk; // clock input
input async_reset; // asynchronous reset low level
output reg Q; // output Q
always @(negedge clk or negedge async_reset)
begin
 if(async_reset==1'b0)
  Q <= 1'b0;
 else
  Q <= D;
end
endmodule
```

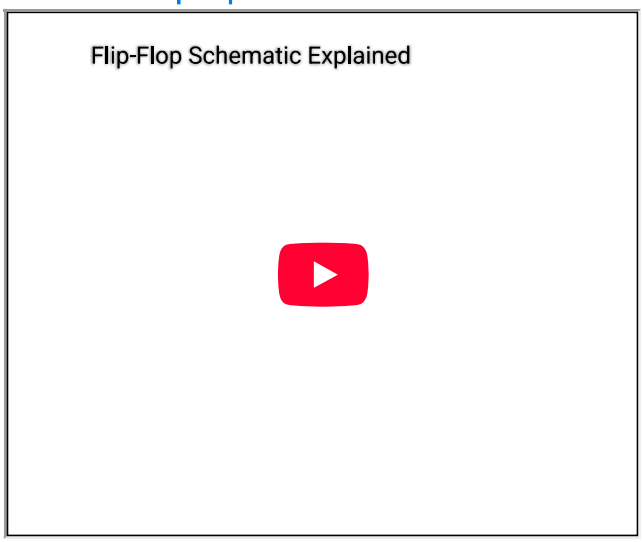
Verilog Testbench code to simulate and verify D Flip-Flop:

```
`timescale 1ns/1ps;
// FPGA projects using Verilog/ VHDL
// fpga4student.com
// Verilog code for D Flip FLop
// Testbench Verilog code for verification
module tb_DFF();
reg D;
reg clk;
reg reset;
wire Q;

RisingEdge_DFlipFlop_SyncReset dut(D,clk,reset,Q);

initial begin
  clk=0;
     forever #10 clk = ~clk;
end
initial begin
 reset=1;
 D <= 0;
 #100;
 reset=0;
 D <= 1;
 #100;
 D <= 0;
 #100;
 D <= 1;
end
endmodule
```

Simulation Waveform for D Flip-Flop:



This tutorial explains in detail the flip-flop schematic and how it works: HERE.



To understand more about setup and hold time of flip-flop, check it out HERE.

Recommended Verilog projects:

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2. Verilog code for FIFO memory
3. Verilog code for 16-bit single-cycle MIPS processor
4. Programmable Digital Delay Timer in Verilog HDL
5. Verilog code for basic logic components in digital circuits
6. Verilog code for 32-bit Unsigned Divider
7. Verilog code for Fixed-Point Matrix Multiplication
8. Plate License Recognition in Verilog HDL
9. Verilog code for Carry-Look-Ahead Multiplier
10. Verilog code for a Microcontroller
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12. Verilog code for Car Parking System
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14. How to load a text file into FPGA using Verilog HDL
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17. Verilog code for comparator design
18. Verilog code for D Flip Flop
19. Verilog code for Full Adder
20. Verilog code for counter with testbench
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23. How to write Verilog Testbench for bidirectional/ inout ports
24. Tic Tac Toe Game in Verilog and LogiSim
25. 32-bit 5-stage Pipelined MIPS Processor in Verilog (Part-1)
26. 32-bit 5-stage Pipelined MIPS Processor in Verilog (Part-2)
27. 32-bit 5-stage Pipelined MIPS Processor in Verilog (Part-3)
28. Verilog code for Decoder
29. Verilog code for Multiplexers
30. N-bit Adder Design in Verilog
31. Verilog vs VHDL: Explain by Examples
32. Verilog code for Clock divider on FPGA

33. How to generate a clock enable signal in Verilog
34. Verilog code for PWM Generator
35. Verilog coding vs Software Programming
36. Verilog code for Moore FSM Sequence Detector
37. Verilog code for 7-segment display controller on Basys 3 FPGA

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